

Homework 5

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Problem 9

Problem 9.1. Let G be a group with normal subgroups $N, H \subseteq G$ such that $N \cap H \cong 0$. Show that $NH \cong N \times H$.

Lemma 9.1. Let G be a group and $H, K \subseteq G$ subgroups. Then HK is a group if and only if $HK = KH$.

Proof. This proof is modified from the proof we did in MA541 homework 6 problem 3 last fall.

Let HK be a group, and let $kh \in KH$ with $k \in K$ and $h \in H$. It follows that $hk \in HK$, and in particular $h^{-1}k^{-1} \in HK$ since H, K are groups, and thus closed under inverse. Since HK is a group, there is some $ab \in HK$ with $a \in H$ and $b \in K$ such that $(h^{-1}k^{-1})^{-1} = ab$. However, since $HK \hookrightarrow G$ is a group homomorphism, we also see that $(h^{-1}k^{-1})^{-1} = kh$, and thus $kh = ab$. Thus, $KH \subseteq HK$. For the converse, let $hk \in HK$, and thus $(hk)^{-1} = ab$ for some $a \in H$ and $b \in K$. It follows that $hk = (ab)^{-1} = b^{-1}a^{-1} \in KH$, and thus $hk \in KH$. The statement follows.

For the other direction, let $HK = KH$. First we show closure. Let $h_1k_1, h_2k_2 \in HK$. Since $HK = KH$, there is thus some $k_3h_3 \in KH$ such that $h_2k_2 = k_3h_3$, giving

$$h_1k_1h_2k_2 = h_1k_1k_3h_3.$$

Since $k_1k_3h_3 \in KH$, there is some $h_4k_4 \in HK$ with $h_4k_4 = k_1k_3h_3$. We thus have

$$h_1k_1h_2k_2 = h_1k_1k_3h_3 = h_1h_4k_4 \in HK.$$

Since $h_1h_4k_4 \in HK$, we have that HK is closed under the group operation. Associativity follows immediately since H, K are subgroups of G , and since $e \in H, K$, we have $e = ee \in HK$. To show inverses exist, let $hk \in HK$, and thus $(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$. Therefore, HK is closed under inverses, meaning HK is a group. ■

Corollary 9.2. Let $H, K \subseteq G$ be subgroups with H normal. Then HK is a subgroup of G .

Proof. Note that if $hk \in HK$ with $h \in H$ and $k \in K$, then $hk \in Hk$. Similarly, $kh \in kH$. Thus,

$$HK = \bigcup_{k \in K} Hk, \quad KH = \bigcup_{k \in K} kH.$$

Since H is normal, for all $k \in K$ we have $Hk = kH$. It follows that

$$HK = \bigcup_{k \in K} Hk = \bigcup_{k \in K} kH = KH.$$

By lemma 9.1, HK is a subgroup of G . ■

Lemma 9.3. Let $H, K \subseteq G$ be subgroups with $H \cap K \cong 0$. Then every $g \in HK$ has a unique representation as $g = hk$ for $h \in H$ and $k \in K$.

Proof. Existence follows by $g \in HK$, so we show uniqueness. Suppose that $g = h_1k_1 = h_2k_2$ are two representations of g in HK . Then

$$k_1 = h_1^{-1}h_2k_2 \implies k_1k_2^{-1} = h_1^{-1}h_2.$$

Since $k_1k_2^{-1} \in K$ and $h_1^{-1}h_2 \in H$, by hypothesis we have $h_1^{-1}h_2 = e$ and $k_1k_2^{-1} = e$. It follows that $h_1 = h_2$ and $k_1 = k_2$. ■

Lemma 9.4. *Let $H, K \subseteq G$ be normal subgroups with $H \cap K \cong 0$. Then for all $h \in H$ and $k \in K$, we have $hk = kh$.*

Proof. Let $h \in H$ and $k \in K$. Consider the commutator

$$[h, k] = hkh^{-1}k^{-1}.$$

Since H is normal, we have $kh^{-1}k^{-1} \in H$. Since K is normal, we have $hkh^{-1} \in K$. Since H, K are closed, we have

$$(hkh^{-1})k^{-1} \in K, \quad h(kh^{-1}k^{-1}) \in H.$$

We thus have

$$hkh^{-1}k^{-1} \in H \cap K = \{e\} \implies hkh^{-1}k^{-1} = e.$$

Therefore, $hk = kh$. ■

Proof of problem 9.1. Since H is normal, by corollary 9.2, NH is a subgroup of G . By lemma 9.3, each $g \in NH$ has a unique representation as $g = nh$ for $n \in N$ and $h \in H$. Define the map $\varphi : NH \rightarrow N \times H$ by $nh \mapsto (n, h)$. The map is well-defined by lemma 9.3, and is manifestly surjective since for all $(n, h) \in N \times H$, there is $nh \in NH$ with $nh \mapsto (n, h)$. Injectivity is also manifest, since if $nh \neq n'h'$, then at least one of $n \neq n'$ or $h \neq h'$ must be true, and thus $(n, h) \neq (n', h')$. It suffices to show φ is a group homomorphism.

The statement will now follow. Let $n_1h_1, n_2h_2 \in NH$. Then

$$\varphi(n_1h_1)\varphi(n_2h_2) = (n_1, h_1) \cdot (n_2, h_2) = (n_1n_2, h_1h_2).$$

By lemma 9.4, we see that $h_1n_2 = n_2h_1$. We thus have

$$\varphi(n_1h_1n_2h_2) = \varphi(n_1n_2h_1h_2) = (n_1n_2, h_1h_2).$$

Therefore, φ is a group homomorphism. ■